AE 598 Q35 Part C

Nitya Jagadam

So we know mean ROE to Hill mapping gives

$$\frac{x}{a} = -\delta e \cos(u-\varphi)$$

$$\frac{y}{a} = 2\delta e \sin(u-\varphi)$$

$$\frac{z}{a} = \delta i \sin(u-\vartheta)$$

Let $A = a\delta e$, $\psi = u-\varphi$

given that $\vartheta = \varphi \pm \pi/2$

$$\sin(u-\vartheta) = \pm \cos\psi$$

$$\therefore \quad x = -A\cos\psi \quad \text{——} (1)$$

$$y = 2A\sin\psi \quad \text{——} (2)$$

$$z = \pm a\delta i \cos\psi \quad \text{——} (3)$$

S/C → spacecraft

(a) constant inter-spacecraft distance

given, $\delta i = \sqrt{3}\,\delta e$

(b) becomes

$$z = \pm \sqrt{3} A \cos\psi$$

Intra s/c distance is

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\Rightarrow \quad \rho^2 = A^2\cos^2\psi + 4A^2\sin^2\psi + 3A^2\cos^2\psi$$

$$= 4A^2(\cos^2\psi + \sin^2\psi)$$

$$= 4A^2$$

$$\Rightarrow \boxed{\rho = 2A = 2a\delta e = \text{constant}}$$

∴ Deputy maintains a constant distance $2a\delta e$ from the chief, producing the circular formation

(b) Tilt of the relative orbital plane

From (a),

$$x = -A\cos\psi \quad , \quad z = \pm\sqrt{3}A\cos\psi$$

$\Rightarrow$ Relative orbit lies in the plane $z = \pm\sqrt{3}x$

Normal vector to this plane: $n_{\text{relative orbit}} = \begin{bmatrix} -\sqrt{3} \\ 0 \\ 1 \end{bmatrix}$

$\Rightarrow$ along track / cross track (y–z) plane is defined by $x = 0$ with normal vector

$$n_{yz} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Let angle γ between the planes is the acute angle between their normals

$$\cos\gamma = \frac{|\bar{n}_{RO} \cdot \bar{n}_{yz}|}{\|\bar{n}_{RO}\| \, \|\bar{n}_{yz}\|}$$

RO → relative orbit

Subbing,

$$\cos\gamma = \frac{\sqrt{3}}{\sqrt{3}+1} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \boxed{\gamma = 30^\circ}$$

∴ The circular relative orbit plane is tilted by exactly 30° from the x-y plane.

(c) Projected circular formation

Now considering $\delta i = 2\delta e$

we already have (1), (2) & (3)

To determine projection onto x-y plane

$$y^2 + z^2 = 4A^2 \sin^2\psi + 4A\cos^2\psi = 4A^2$$

$$\Rightarrow y^2 + z^2 = (2a\delta e)^2$$

this is a circle of radius

$$\boxed{R_{projected} = 2a\delta e}$$

To determine the tilt of the actual orbital plane observe that

$$z = \pm 2x$$

normal vector to relative orbit plane is

$$n_{\text{relative orbit}} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

Using the normal to y-z plane

$$\cos\gamma = \frac{|\bar{n}_{RO} \cdot \bar{n}_{yz}|}{\|\bar{n}_{RO}\| \, \|\bar{n}_{yz}\|} = \frac{2}{\sqrt{5}}$$

$$\Rightarrow \boxed{\gamma \approx 26.57^\circ}$$

∴ The projected circular formation has a circular y-z projection while its actual orbital plane is tilted by approx 26.57°

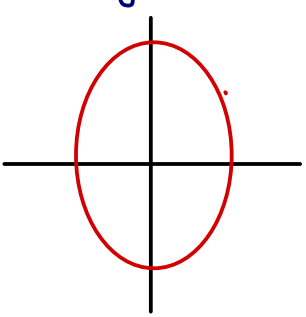
(d) Mission safety

Earth pointing distributed imaging mission

↳ choosing the projected-circular formation makes sense because its projection onto the along-track/cross-track (y-z) plane is circular.

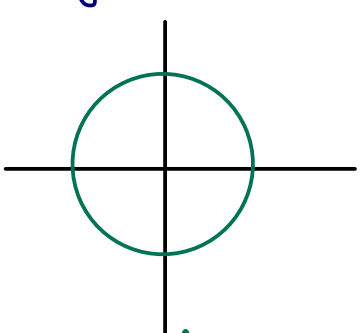
projected circular formulation → $\delta i = 2\delta e$

x-y view



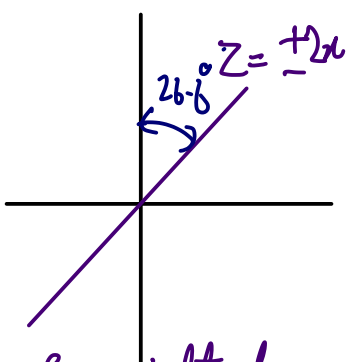
2:1 ellipse

y-z view



circular

x-z view

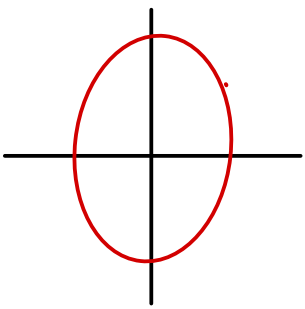


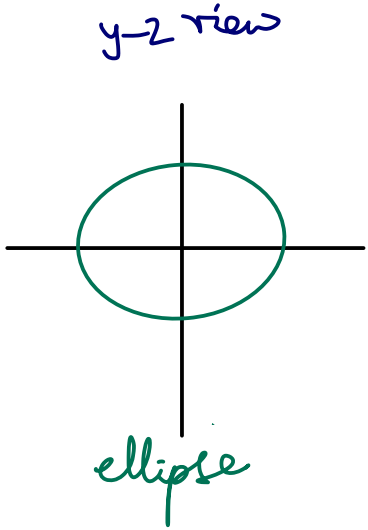
plane tilt line

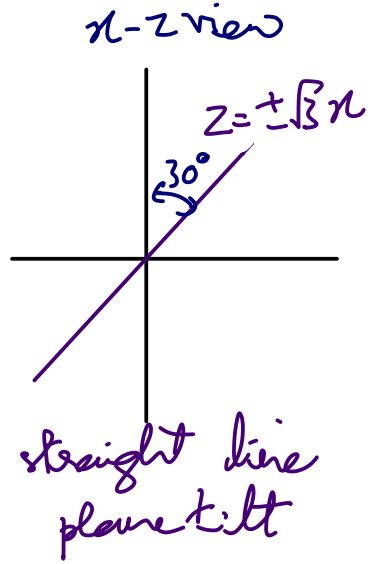
Interferometric or distributed telescope mission:

↳ Choosing the circular formation makes sense because it maintains a constant 3D-inter s/c distance.

circular formation → $\delta i = \sqrt{3}\,\delta e$







Passive safety

Passive collision safety under along-track uncertainty depends on maintaining radial-cross-track separation:

$$\delta r_{xz} = \sqrt{x^2 + z^2}$$

For both formations,

$$\vartheta = \varphi \pm \frac{\pi}{2}$$

⇒ the relative eccentricity & inclination vectors are orthogonal:

$$\delta e \cdot \delta i = \delta e \, \delta i \cos(\varphi - \vartheta) = 0$$

We know passive safety criterion is

$$\delta q_{xz,min} = 0$$

At certain orbital phases, $x \& z$ vanish simultaneously. If along track position becomes uncertain, the nominal along track separation cannot guarantee collision avoidance.

∴ Neither formation is passively safe against along-track uncertainty despite nonzero 3D separation of the circular formulation.